Subiect 1

Florea
Daria
Mihaela

Factorizările $A = LDL^T$ și $A = LL^T$

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Aplicăm factorizarea Cholesky

$$A = L \times L^T = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \times \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ -1/\sqrt{2} & \sqrt{3}/\sqrt{2} & 0 \\ 0 & -\sqrt{2}/\sqrt{3} & 2/\sqrt{3} \end{bmatrix} \times \begin{bmatrix} \sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & \sqrt{3}/\sqrt{2} & -\sqrt{2}/\sqrt{3} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}$$

$l_{11}^2 = a_{11} \iff l_{11}^2 = 2 \iff l_{11} = \sqrt{2}$

$l_{11} l_{21} = a_{12} \iff \sqrt{2} \cdot l_{21} = -1 \iff l_{21} = \frac{-1}{\sqrt{2}}$

$\sqrt{2} \cdot l_{31} = 0 \Rightarrow l_{31} = 0$

$\left(-\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{2}}\right) + l_{22}^2 = 2 \iff \frac{1}{2} + l_{22}^2 = 2 \iff$

$l_{22}^2 = \frac{3}{2} \iff l_{22} = \sqrt{\frac{3}{2}}$

$\frac{\sqrt{3}}{\sqrt{2}} \cdot l_{32} = -1 \iff l_{32} = \frac{-\sqrt{2}}{\sqrt{3}}$

$\left(\frac{-\sqrt{2}}{\sqrt{3}}\right)^2 + l_{33}^2 = 2 \iff \frac{2}{3} + l_{33}^2 = 2 \iff$

$l_{33}^2 = \frac{4}{3} \iff l_{33} = \frac{2}{\sqrt{3}}$ (alegem soluția pozitivă)

$$\Rightarrow \quad L = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ -1/\sqrt{2} & \sqrt{3}/\sqrt{2} & 0 \\ 0 & -\sqrt{2}/\sqrt{3} & 2/\sqrt{3} \end{bmatrix} \approx \begin{bmatrix} 1,4142 & 0 & 0 \\ -0,7071 & 1,2247 & 0 \\ 0 & -0,8165 & 1,1547 \end{bmatrix}$$

Pentru factorizarea $A = LDL^T$

alegem $D = I_3$ (matrice diagonală) și

obținem:

$$A = \begin{bmatrix} 1,4142 & 0 & 0 \\ -0,7071 & 1,2247 & 0 \\ 0 & -0,8165 & 1,1547 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot$$

$$\begin{bmatrix} 1,4142 & -0,7071 & 0 \\ 0 & 1,2247 & -0,8165 \\ 0 & 0 & 1,1547 \end{bmatrix}$$